

RITIKESH KUMAR SINGH

Vadodara, Gujarat | +91 87893-89286 | rksingh108@outlook.com
rksingh-108.github.io | github.com/rksingh-108 | linkedin.com/in/rksingh108

PROFESSIONAL SUMMARY

BCA student and software developer with hands-on experience building full-stack web applications, backend APIs, database-driven systems, and AI-powered products—taken from development through cloud deployment. Practical foundation in real-time systems, authentication, and relational databases, with growing expertise in LLMs, agentic workflows, and retrieval-augmented generation. Open to Software Development, Full-Stack, Backend, or AI/LLM Engineering internships.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript/JavaScript, React, Next.js, Flask, Express.js, Fastify

Databases: PostgreSQL, pgvector, MongoDB, SQLite

AI & LLM: Groq, Retrieval-Augmented Generation (RAG), Agentic Tool-Calling, LLM Integration

Cloud & DevOps: Docker, Vercel, Render, Supabase, Azure, Git/GitHub

Core Competencies: REST APIs, JWT/RBAC Authentication, Real-Time Systems, Database Design

PROJECTS

TRYLO — Full-Stack Ride Booking Platform

Next.js, React, Express.js, PostgreSQL, Prisma, Socket.IO, MapLibre, Docker, Azure

- Built a Turborepo/pnpm monorepo with separate Customer, Driver, and Admin apps sharing a common backend, covering the full ride lifecycle from booking through completion
- Designed a geo-indexed driver-matching engine using PostgreSQL cube/earthdistance with a GiST-indexed location column and a progressive 1km → 2km → 3km search radius
- Implemented real-time driver location tracking over Socket.IO, streaming live GPS updates to customers throughout active rides
- Built phone + OTP authentication with JWT access/refresh tokens and role-based access control across customer, driver, and admin roles
- Developed an internal wallet-based payment system with atomic, idempotent settlement logic, plus an admin dashboard for managing customers, drivers, rides, and transactions
- Integrated MapLibre GL JS, OSRM routing, and Nominatim/Photon geocoding for live maps and navigation; containerized with Docker and deployed on Azure (Container Apps, Static Web Apps, PostgreSQL Flexible Server)

WorkPilot AI — AI-Powered Workplace Assistant

Next.js, TypeScript, Fastify, PostgreSQL, pgvector, Groq, RAG, JWT/RBAC, Vercel, Render, Supabase

- Built and deployed an AI agent that unifies tasks, emails, calendar, meetings, and company knowledge into personalized daily plans and meeting preparation via agentic tool-calling and multi-step workflows
- Implemented RAG-based company knowledge search using PostgreSQL + pgvector to ground agent responses in relevant internal documents
- Built authentication, role-based access control, human-in-the-loop approval workflows, and full activity/audit logging for safer agent actions
- Deployed and operated the system in production (Vercel, Render, Supabase), addressing real-world concerns including LLM rate limits, timezone-correct scheduling, and safe response rendering

Web-Based Employee Performance Tracker

Flask, SQLAlchemy, SQLite, Chart.js

- Built a role-based (admin/employee) performance management system with session-based authentication, hashed credentials, and protected routes
- Designed a relational schema (Departments, Employees, Users, Performance) with full CRUD via the Flask-SQLAlchemy ORM, and a multi-factor performance scoring engine (attendance, task completion, teamwork, communication, project rating) with automatic grade classification
- Built an analytics dashboard with Chart.js visualizations for department-wise performance and grade distribution, plus individual employee and department reports

EDUCATION

Bachelor of Computer Applications (BCA)

Parul University, Vadodara

Expected Graduation: 2027 | 5th Semester | CGPA: 7.81/10